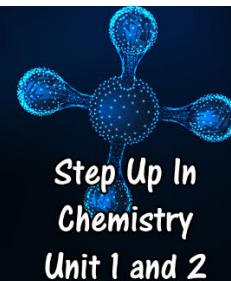


2.4 Covalent Bonding

Problems Worksheet



1. Explain how a covalent bonds forms.
2. Discuss the difference between a single, double and triple covalent bond. Give an example of a molecule for each.
3. Covalent bonds generally only form between non-metal atoms. Why is this?

4. Explain the difference between covalent molecules and covalent networks. Why are their melting and boiling points so different?

5. Draw Lewis diagrams for the following molecules.

a. H_2

g. CO_2

b. Cl_2

h. NH_3

c. N_2

i. SO_2

d. O_2

j. SO_3

e. HF

k. PCl_3

f. H_2O

6. Polyatomic ions can form ionic compounds with other ions. However, polyatomic ions are held together by covalent bonds. Draw Lewis diagrams for the following ions:



7. Explain the following properties of covalent molecular substances:

a. Electrical conductivity

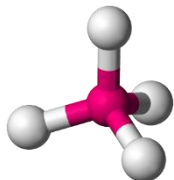
b. Melting and boiling points

8. Name the following shapes of covalent molecules.

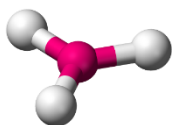
a.



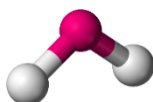
b.



c.



d.



e.

